iRAYPLE Acquisition Card

SDK Development Document (LabView)



V1.1.0

Revision History

No.	Date	Version	Revision Content	SDK Version
1	Jul. 2023	1.0.0	First Release.	2.4.0
2	Mar. 2024	1.1.0	Added the VI Obtainment Parameter of Bool and String Attribute Setting.	2.4.2

401674 Iraypie 2025-01-18

Table of Contents

1 Overview 1

1.1 Download 1

1.2 Compatibility Introduction 1

1.3 Installation Directory 1

1.3.1 SDK for Windows 2

1.4 Related Resources of SDK (C interface) 3

2 Calling Process and Software Introduction 4

2.1 Overall Calling Process of SDK 4

2.2 Software Introduction 5

3 Sub-VI Function Introduction 8

3.1 EnumInterface.vi 8

3.2 EnumDevice.vi 8

3.3 OpenInterface.vi 9

3.4 OpenDevice.vi 9

3.5 GetIntFeatureValue.vi 10

3.6 GetDoubleFeatureValue.vi 10

3.7 GetEnumFeatureValue.vi 10

3.8 GetEnumFeatureEntryNum.vi 11

3.9 GetEnumFeatureEntrys.vi 11

3.10 GetEnumFeatureSymbol.vi 11

3.11 SetIntFeatureValue.vi 12

3.12 SetDoubleFeatureValue.vi 12

3.13 SetEnumFeatureSymbol.vi 12

3.14 IsOpenInterface.vi 13

3.15 IsOpenDevice.vi 13

3.16 GetFrame.vi 13

3.17 ReleaseFrame.vi 14

AOI Iraple 2025-01-10

3.18 ExecuteCommandFeature.vi 14
3.19 PixelConvert.vi 15
3.20 SaveImageToBmp.vi 15
3.21 StartGrabbing.vi 15
3.22 StopGrabbing.vi 16
3.23 CloseDevice.vi 16
3.24 CloseInterface.vi 16
3.25 SetBufferCount.vi 17
3.26 IsGrabbing.vi 17
3.27 SetBoolFeatureValue.vi 17
3.28 SetStringFeatureValue.vi 18
3.29 GetBoolFeatureValue.vi 18
3.30 GetStringFeatureValue.vi 18
4 FAQ (Frequently Asked Question) 20

401674 Iraypl e 2025-01-18

1 Overview

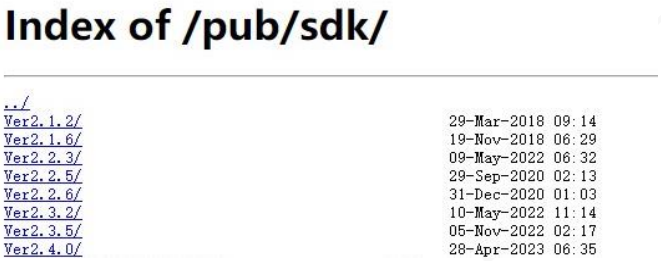
1.1 Download

You can download the SDK installation package on the official website:

http://download.huaraytech.com/pub/sdk/

Please download the appropriate version of SDK.

Figure 1-1 SDK installation packages



1.2 Compatibility Introduction

Table 1-1 Operating system requirement

OS Type	Version	Note
Windows *1	Win7 SP1 32-bit/64-bit	Ensure that OS version includes the patch of SP1.
	Win8 32-bit/64-bit	
	Win10 32-bit/64-bit	
Linux (x86) *2	**System Version** Ubuntu 12.04 LTS ~ Ubuntu 21.04 LTS (32-bit/64-bit)	The upgraded kernel version system is excluded.
	CentOS 6.5~CentOS 32 (32-bit/64-bit)	

- Windows XP is no longer supported since SDK V2.2.5.
- The supported kernel version of Linux is 2.6.32 to 5.11.0 (Inclusive).
- Use the command of **uname –a** to check the kernel version number.

1.3 Installation Directory

1.3.1 SDK for Windows

The installation directory structure of the SDK for Windows is as follows:

Figure 1-2 installation directory



Since the 2.1.0 version of SDK, user can select the installation path.

Table 1-2 Installation directory structure description

Application	● **MViewer** is compatible with 32-bit and 64-bit. ● **CamTools** is compatible with 32-bit and 64-bit.
Development	**3rdPartyPlatformAdapter** It includes two plug-in dynamic libraries of third-party algorithm platform (Halcon and Sherlock).
	DOTENT DLL It includes **.Net** library file which is imported when user create the project.
	Include It includes the header file of the SDK interface, which explains the basic attributes of various interfaces. The specific usage of header files is in the **samples folder**. The other is the extension definition of the header file of the SDK, which defines the SDK error code, the user-defined parameters, the data restriction of some parameters, the enumeration of various behaviors, and the data structure of various information.
	Lib It includes the lib files.
	Samples It includes the demo files (C#, Delphi, Lab View, MFC, Python, QT, VB, VC).
Documentations	● **.Net** interface document ● **ActiveX** interface document

401674 Private 2025-01-18

	● **C** interface document ● **SDK** development document
Drivers	● The installation and uninstallation software of **DShowFilter** high-performance driver. ● The installation and uninstallation software of **GigE** high-performance driver. ● The installation and uninstallation software of **PCIeCL** high-performance driver. ● The installation and uninstallation software of **USB** high-performance driver. ● The installation and uninstallation software of **CLVSP** high-performance driver.
Runtime	The compilation dynamic link library when running the SDK. It includes the third-party **GenICam** library which is separated with two folders as 32-bit and 64-bit.

1.4 Related Resources of SDK (C interface)

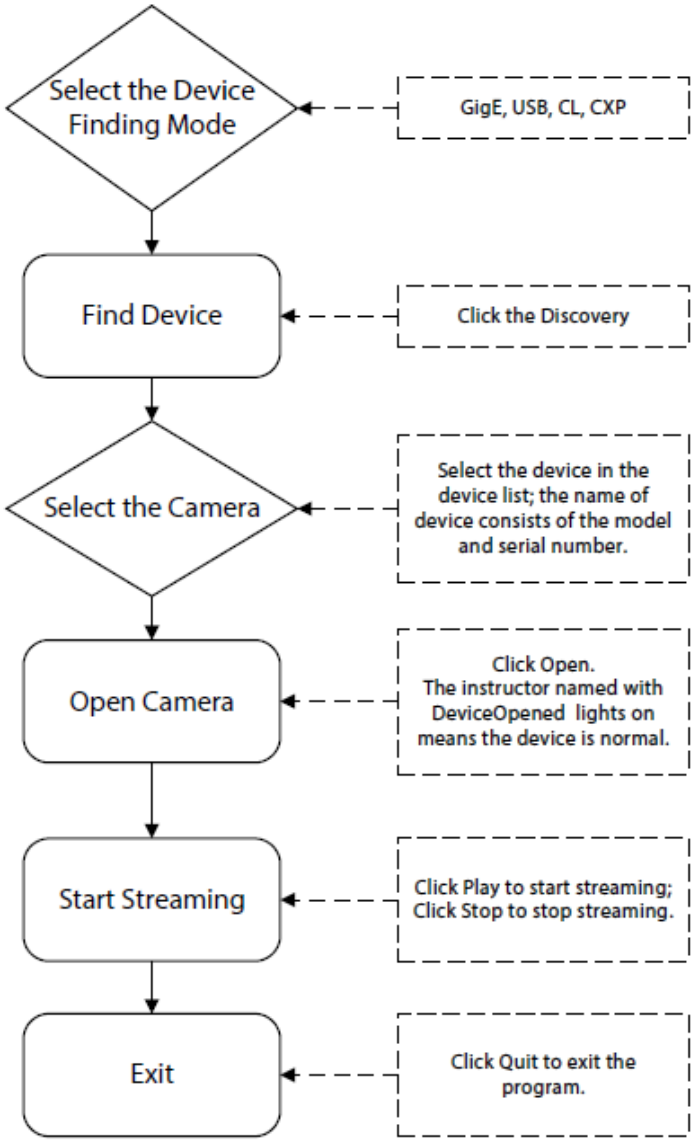
Category	Item	File Storage Folder
SDK Library	Header File	【SDK Installation Directory】\Development\Include\IMVApi.h \Development\Include\IMVDefines.h
	Lib (32-bit)	【SDK Installation Directory】\Development\Lib\win32\MVSDKmd.lib
	Lib (64-bit)	【SDK Installation Directory】\Development\Lib\x64\MVSDKmd.lib
	Runtime Library (32-bit)	【SDK Installation Directory】\Runtime\Win32\MVSDKmd.dll
	Runtime Library (64-bit)	【SDK Installation Directory】\Runtime\x64\MVSDKmd.dll
C API Document	Chinese	【SDK Installation Directory】\ Documentations\ GenICam_API_C_CHS.chm
	English	【SDK Installation Directory】\ Documentations\ GenICam_API_C_ENG.chm

2 Calling Process and Software Introduction

The LabView interface of acquisition card routine includes single-camera routine and dual-camera routine.

2.1 Overall Calling Process of SDK

Figure 2-1 SDK calling process



2.2 Software Introduction

Figure 2-2 LabView main interface

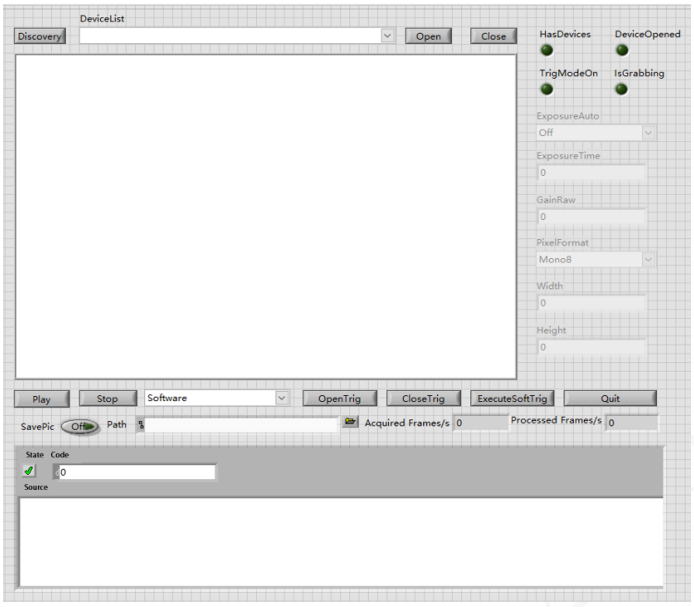
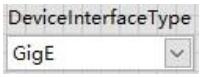
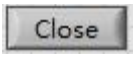
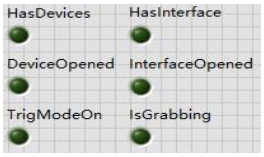
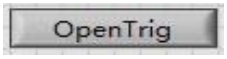
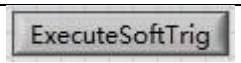
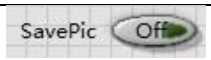
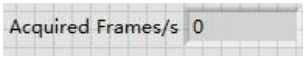
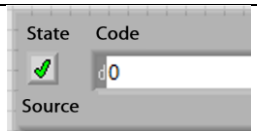


Table 2-1 LabView function description

No.	Diagram	Description
1		The methods of finding acquisition card and camera includes **GigE**, **U3v**, **CL** and **CXP**.
2		After clicking the discovery, wait for the instructor of **HasDevice** and **HasInterface** to light up. If LabView is in the status of “Not Respond” during device finding, please do not close the software.
3		The device list will only display the acquisition card which has been connected with the camera. The name of device in the list consists of device model and serial number.
4		After clicking the open button, the selected device in the device list will be opened, and the instructor lights of **DeviceOpened** and **InterfaceOpened** will light up.
5		After closing the device, the instructor of the **DeviceOpened** and **InterfaceOpened** will go off.

6		• HasDevice: The instructor of find camera. • HasInterface: The instructor light of find acquisition card. • DeviceOpened: The instructor light of open camera. • InterfaceOpened: The instructor light of open acquisition card. • TrigModeOn: The instructor of enable trigger mode. • IsGrabbing: The instructor light of start streaming.
7		To select the camera type, such as line scan camera and area scan camera.
8		To select the pixel format of the camera and acquisition card. The setting will be synchronized into acquisition card and camera simultaneously.
9		The drop-down list of **ExposureAuto** includes Off, Once and Continuous.
10		To configure the width and height parameters of the camera and acquisition card. The settings will be synchronized into acquisition card and camera simultaneously.
11		To configure the exposure event and gain value.
12		Start streaming. After clicking this button, the instructor light of **IsGrabbing** will light up.
13		Stop streaming. After clicking this button, the instructor light of **IsGrabbing** will go off.
14		To select the supported trigger modes.
15		To configure the selected trigger mode. After completing the configuration, the instructor light of **TrigModeOn** will light up.
16		Disable the Trigger mode. After disabling, the instructor of **TrigModeOn** will go off.
17		Execute the software triggering.
18		To discontinue the operation and release the handles of camera and acquisition card.

19		To select the image saving path. The image name will be the **blockID**.
20		To enable or disable the image saving function.
21		Acquired Frames/s: It displays whether the frame of image is required. If it is not 0, the image is required; otherwise, the image is not required.
22		Displays the operation error information.

401674 Irayple 2025-01-18

3 Sub-VI Function Introduction

- Library file path: LabView\Depend.
- Executable file path: LabView\bin.
- The directory of interface header files of SDK: \Development\Include\IMVApi.h
- The directory of the macro definition, enumeration, struct and callback function statement of SDK interface: \Development\Include\IMVDefines.h
- Sub-VI is the encapsulation of SDK interface, and the corresponding sub-VI can be introduced into the project to realize the corresponding functions.

3.1 EnumInterface.vi

- Interface Description: Acquisition card enumeration.

Table 3-1 Parameter description

Parameter	Description
nInterfaceType [IN]	Enumeration device interface type (default type: ALL)
pInterfaceList [OUT]	The acquisition card list.
nDevNum [OUT]	The number of the found acquisition card.

- Function Description:
  - ◇ After calling **IMV_FG_VI_EnumInterface**, the interface obtains the acquired image quantity and store the acquisition card information into the memory by coding the parameters of **nInterfaceType**.
  - ◇ Then, the interface obtains the device information of acquisition card and transmits it into the LabView by calling the **IMV_FG_VI_Assignment**.

3.2 EnumDevice.vi

- Interface Description: Camera enumeration.

Table 3-2 Parameter description

Parameter	Description
nDeviceType [IN]	The interface type pf enumeration device.
deviceKeyList [OUT]	Acquisition card list.
nDeviceNum [OUT]	The number of the found device.

- Function Description:

  ◇ After calling **IMV_FG_VI_EnumDevices**, obtain the number of devices and store the information of the camera and acquisition card into the memory by adding the **nDeviceType**.

  ◇ Then, obtain and transmit the device list into the LabView by calling the **IMV_FG_VI_DeviceAssignment**.

3.3 OpenInterface.vi

- Interface Description: Open acquisition card.

Table 3-3 Parameter description

Parameter	Description
nIndex [IN]	The index of acquisition card.
hIFDev [OUT]	The handle of acquisition card.
return type [OUT]	The return value of function.

- Function Description: After calling the **IMV_FG_ VI_OpenInterface** of the SDK interface, open acquisition card by adding the **nIndex**.

3.4 OpenDevice.vi

- Interface Description: Open camera.

Table 3-4 Parameter description

Parameter	Description
mode [IN]	Open camera mode.
pIdentifier [IN]	Specify the identifier (device key, custom device name, explicitly casting the pointer representing the IP address from char type to void type; explicitly casting the pointer representing the index from unsigned int type to void type.)
hDev [OUT]	The camera handles.
return type [OUT]	The return value of function.

- Function Description: After calling the **IMV_VI_SetBoolFeatureValue** of the SDK interface, open the camera by transmitting the mode.

3.5 GetIntFeatureValue.vi

- Interface Description: Obtain the attribute value of Int.

Table 3-5 Parameter description

Parameter	Description
handle [IN]	The handles of cameras and acquisition card.
pFeatureName [IN]	The obtained attribute name.
pIntValue [OUT]	The attribute value of int.

- Function Description: After calling the **IMV_FG_ VI_GetIntFeatureValue** of the SDK interface, obtain the corresponding attribute value by adding the handle and **pFeatureName**.

3.6 GetDoubleFeatureValue.vi

- Interface Description: Obtain the attribute value of Double.

Table 3-6 Parameter description

Parameter	Description
handle [IN]	The handles of cameras and acquisition card.
pFeatureName [IN]	The obtained attribute name.
pDoubleValue [OUT]	The attribute value of Double.

- Function Description: After calling the **IMV_FG_ VI_GetDoubleFeatureValue**, obtain the corresponding attribute value by adding the **handle** and **pFeatureName**.

3.7 GetEnumFeatureValue.vi

- Interface Description: Obtain the attribute value of Enum.

Table 3-7 Parameter description

Parameter	Description
handle [IN]	The handles of cameras and acquisition card.
pFeatureName [IN]	The obtained attribute name.
pEnumValue [OUT]	The Enum attribute value.

- Function Description: After calling the **IMV_FG_ VI_GetEnumFeatureValue**, obtain the corresponding attribute value by adding the **handle** and **pFeatureName**.

3.8 GetEnumFeatureEntryNum.vi

- Interface Description: Obtain the number of configurable enumeration values of enumeration attribute.

Table 3-8 Parameter description

Parameter	Description
handle [IN]	The handles of cameras and acquisition card.
pFeatureName [IN]	The obtained attribute name.
pEntryNum [OUT]	Obtain the number of configurable enumeration values of enumeration attribute.

- Function Description: After calling the **IMV_FG_ VI_GetEnumFeatureEntryNum**, obtain the number of configurable enumeration value by adding the **handle** and **pFeatureName**.

3.9 GetEnumFeatureEntrys.vi

- Interface Description: Obtain the list of configurable enumeration values of enumeration attribute.

Table 3-9 Parameter description

Parameter	Description
handle [IN]	The handles of cameras and acquisition card.
EntryNum [IN]	The number of configurable enumeration values of enumeration attribute.
pFeatureName [IN]	The obtained attribute name.
pLbEnumEntryList [out]	The number of configurable enumeration values of enumeration attribute.

- Function Description: After calling the **IMV_FG_ VI_GetEnumFeatureEntrys**, obtain the **pLbEnumEntryList** by adding the **handle**, **EntryNum** and **pFeatureName**.

3.10 GetEnumFeatureSymbol.vi

- Interface Description: Obtain the enumeration symbol value.

Table 3-10 Parameter description

Parameter	Description
handle [IN]	The handles of cameras and acquisition card.
pFeatureName [IN]	The obtained attribute name.

pEnumSymbol [OUT]	The symbol value of enumeration attribute.

- Function Description: After calling the **IMV_FG_ VI_GetEnumFeatureSymbol**, obtain the desired symbol value of attribute value by adding the **handle** and **pFeatureName**.

3.11 SetIntFeatureValue.vi

- Interface Description: Set the attribute value of Int.

Table 3-11 Parameter description

Parameter	Description
handle [IN]	The handles of cameras and acquisition card.
pFeatureName [IN]	The configured attribute name.
intValue [IN]	The configured int value.

- Function Description: After calling the **IMV_FG_ VI_SetIntFeatureValue**, configure the value of **intValue** by adding the **handle** and **pFeatureName**.

3.12 SetDoubleFeatureValue.vi

- Interface Description: Configure the attribute value of Double.

Table 3-12 Parameter description

Parameter	Description
handle [IN]	The handles of cameras and acquisition card.
pFeatureName [IN]	The configured attribute name.
doubleValue [IN]	The configured attribute value of Double.

- Function Description: After calling the **IMV_FG_ VI_SetDoubleFeatureValue**, configure the value of **doubleValue** by adding the **handle** and **pFeatureName**.

3.13 SetEnumFeatureSymbol.vi

- Interface Description: Configure the symbol value of the enumeration attribute.

Table 3-13 Parameter description

Parameter	Description
handle [IN]	The handles of cameras and acquisition card.
pFeatureName [IN]	The obtained attribute name.

pEnumSymbol [IN]	The symbol value of enumeration attribute.
return type [OUT]	The return value of function.

- Function Description: After calling the **IMV_FG_ VI_SetEnumFeatureSymbol** of the SDK interface, configure the value of **pEnumSymbol** by adding the **handle** and **pFeatureName**.

3.14 IsOpenInterface.vi

- Interface Description: Determine whether the acquisition card is opened.

Table 3-14 Parameter description

Parameter	Description
interface handle [IN]	The handles of acquisition card.
return type [OUT]	The return value of function.

- Function Description: After calling the **IMV_FG_ VI_IsOpenInterfaceof** the SDK interface, determine whether the acquisition card is opened according to the return value by adding **interface handle**.

3.15 IsOpenDevice.vi

- Interface Description: Determine whether the device is opened.

Table 3-15 Parameter description

Parameter	Description
dev handle [IN]	The camera handles.
return type [OUT]	The return value of function.

- Function Description: After calling the **IMV_FG_ VI_IsDeviceOpen** of the SDK interface, determine whether the device is opened according to the return value by adding **dev handle**.

3.16 GetFrame.vi

- Interface Description: Acquire a frame of image (Synchronous acquisition mechanism for frame data).

Table 3-16 Parameter description

Parameter	Description
interface handle [IN]	The handles of acquisition card.

timeoutMS [IN]	Timeout duration.
pLbFrame out [OUT]	The frame data information.
pData [OUT]	The starting address of memory of the fame data of image.
framehandle [OUT]	The frame handles.
return type [OUT]	The return value of function.

- Function Description: After calling the **IMV_FG_ VI_GetFrame**, obtain the **pData**, **pLbFrame out** and **framehandle** by adding the **timeoutMS** and **interface handle**.

3.17 ReleaseFrame.vi

- Interface Description: Release the image buffer memory.

Table 3-17 Parameter description

Parameter	Description
interface handle [IN]	The handles of acquisition card.
pFrame [IN]	The frame data information.
pData [IN]	The starting address of memory of the fame data of image.
framehandle [IN]	The frame handles.

- Function Description: After calling the **IMV_FG_ VI_ReleaseFrame** of the SDK interface, delete the **pLbFrame**, **interface handle**, **framehandle**, **pData**, etc.

3.18 ExecuteCommandFeature.vi

- Interface Description: Execute the command attribute.

Table 3-18 Parameter description

Parameter	Description
handle [IN]	The handles of cameras and acquisition card.
pFeatureName [IN]	The attribute name.
return type [OUT]	The return value of function.

- Function Description: After calling the **IMV_FG_ VI_ExecuteCommandFeature**, execute the command attribute by adding the **handle** and **pFeatureName**.

3.19 PixelConvert.vi

- Interface Description: Pixel format conversion.

Table 3-19 Parameter description

Parameter	Description
lbPstPixelConvertParam [IN]	The pixel format conversion parameter structure.
Interface handle [IN]	The handles of acquisition card.
pSrcData [IN]	The input image data.
pDstBuf [IN]	The applied output image data buffer.
pDstBuf out [OUT]	The output image data.

- Function Description: After calling the **IMV_FG_ VI_PixelConvert**, obtain the **pDstBuf** out by adding the **lbPstPixelConvertParam**, **Interface handle**, **pSrcData** and **pDstBuf**.

3.20 SaveImageToBmp.vi

- Interface Description: Save image.

Table 3-20 Parameter description

Parameter	Description
Interface handle [IN]	The handles of acquisition card.
pLbFrame [IN]	The frame data information.
filePath [IN]	The inputted file paths.
pDstBuf [IN]	The converted image data.
pSrcData [IN]	The unconverted image data.

- Function Description: After calling the **IMV_FG_ VI_SaveImageToBmp** of the SDK interface, save image into the specified path by adding **Interface handle**, **pLbFrame**, **filePath**, **pDstBuf** and **pSrcData**.

3.21 StartGrabbing.vi

- Interface Description: Start streaming.

Table 3-21 Parameter description

Parameter	Description
interface handle [IN]	The handles of acquisition card.
return type [OUT]	The return value.

- Function Description: After calling the **IMV_FG_ VI_StartGrabbing**, determine whether the streaming is successful according to the return value by adding **interface handle**.

3.22 StopGrabbing.vi

- Interface Description: Stop acquiring image.

Table 3-22 Parameter description

Parameter	Description
interface handle [IN]	The handles of acquisition card.
return type [OUT]	The return value.

- Function Description: After calling the **IMV_FG_ VI_StopGrabbing**, determine whether the streaming is successful according to the return value by adding **interface handle**.

3.23 CloseDevice.vi

- Interface Description: Close device.

Table 3-23 Parameter description

Parameter	Description
dev handle [IN]	The handle of camera.
return type [OUT]	The return value of function

- Function Description: After calling the **IMV_FG_ VI_CloseDevice**, determine whether the device is closed according to the return value by adding **dev handle**.

3.24 CloseInterface.vi

- Interface Description: Close acquisition card.

Table 3-24 Parameter description

Parameter	Description

interaface handle [IN]	The handles of acquisition card.
return type [OUT]	The return value of function.

- Function Description: After calling the **IMV_FG_ VI_CloseInterface**, determine whether the acquisition card is closed according to the return value by adding **return type**.

3.25 SetBufferCount.vi

- Interface Description: Configure the buffer quantity of frame data.

Table 3-25 Parameter description

Parameter	Description
interaface handle [IN]	The handles of acquisition card.
nSize [IN]	The buffer quantity.
return type [OUT]	The return value of function.

- Function Description: After calling the **IMV_FG_ VI_SetBufferCount**, configure the buffer quantity of frame data by adding **interface handle** and **nSize**; Determine whether the acquisition card is closed according to the **return type**.

3.26 IsGrabbing.vi

- Interface Description: Determine whether the acquisition card is streaming.

Table 3-26 Parameter description

Parameter	Description
interaface handle [IN]	The handles of acquisition card.
return type [OUT]	The return value of function.

- Function Description: After calling the **IMV_FG_VI_IsGrabbing**, determine whether the device is streaming according to the return type by adding **interface handle**.

3.27 SetBoolFeatureValue.vi

- Interface Description: Configure the attribute value of Bool.

Table 3-27 Parameter description

Parameter	Description
handle [IN]	The handles of cameras and acquisition card.

pFeatureName [IN]	The configured attribute name.
boolValue [IN]	The configured attribute value of Bool.

- Function Description: After calling the **IMV_FG_ VI_SetBoolFeatureValue**, configure the value of **boolValue** by adding the **handle** and **pFeatureName**.

3.28 SetStringFeatureValue.vi

- Interface Description: Configure the attribute value of String.

Table 3-28 Parameter description

Parameter	Description
handle [IN]	The handles of cameras and acquisition card.
pFeatureName [IN]	The configured attribute name.
pStringValue [IN]	The configured attribute value of String.

- Function Description: After calling the **IMV_FG_ VI_SetStringFeatureValue**, configure the value of **pStringValue** by adding the **handle** and **pFeatureName**.

3.29 GetBoolFeatureValue.vi

- Interface Description: Obtain the attribute value of Bool.

Table 3-29 Parameter description

Parameter	Description
handle [IN]	The handles of cameras and acquisition card.
pFeatureName [IN]	The Obtained attribute name.
pBoolValue [OUT]	The Bool attribute value.
return type [OUT]	The return value.

- Function Description: After calling the **IMV_FG_ VI_GetBoolFeatureValue** of the SDK, obtain the corresponding attribute value by adding the **handle** and **pFeatureName**.

3.30 GetStringFeatureValue.vi

- Interface Description: Obtain the attribute value of String.

Table 3-30 Parameter description

Parameter	Description
handle [IN]	The handles of cameras and acquisition card.
pFeatureName [IN]	The obtained attribute name.
pStringValue [OUT]	The attribute value of Bool.
return type [OUT]	The return value.

- Function Description: After calling the **IMV_FG_ VI_GetStringFeatureValue**, obtain the corresponding attribute value by adding the **handle** and **pFeatureName**.

401674 Iraypl e 2025-01-18

4 FAQ (Frequently Asked Question)

1. When the version of SDK is 2.4.1, the routine cannot be run on the 32LabView. The version of 2.4.2 and later versions 32-bit dll library can solve the problem.

401674 Irayple 2025-01-18